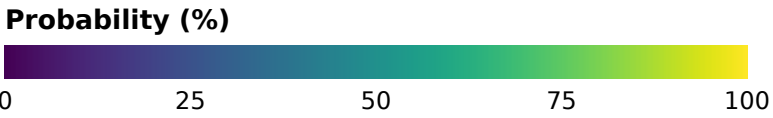


AI-assisted Millennium Problem forecasts

Correct complete solution by deadline (%) · Models ordered by public release date



Birch and Swinnerton-Dyer conjecture		2026	2030	2035	2040	2050	Hodge conjecture		2026	2030	2035	2040	2050	Navier-Stokes existence and smoothness		2026	2030	2035	2040	2050
GPT-4 · 2023-06-13		—	—	—	—	—	—		—	—	—	—	—	—		—	—	—	—	—
Qwen 72B · 2023-11-30		—	—	—	—	—	—		—	—	—	—	—	—		—	—	—	—	—
GPT-4o · 2024-08-06		1	5	15	30	50	1		5	15	30	50	50	2		5	15	30	45	50
Qwen2.5 72B · 2024-09-19		10	25	40	60	80	10		22.5	40	60	80	80	10		20	40	60	75	80
o1 · 2024-12-17		1	3	10	20	37.5	1		3	8	16.5	37.5	37.5	1		4	11	25	40	45
R1 Distill 32B · 2025-01-20		—	—	—	—	—	—		—	—	—	—	—	—		—	—	—	—	—
o3 · 2025-04-16		0.5	2	5	11.5	25	0.5		2	6	12	25	25	1		5	15	28	40	45
Qwen3 32B · 2025-04-29		1	5	10	25	45	1		3	7	15	30	30	2		5	15	30	40	40
GPT-5 · 2025-08-07		0.1	1	3.2	11	24.5	0.1		0.8	3.1	8.8	22	22	0.3		2.5	9	20	37	37
Claude Opus 4.5 · 2025-11-24		0.1	1.2	4.5	10	20	0.1		0.8	3.2	7	15	15	0.2		2	6	12	23	23
Claude Opus 4.6 · 2026-02-05		0.05	0.5	2	6	12	0.02		0.3	2	6	14	14	0.1		1	4	9	18	18
Qwen3.5 27B · 2026-02-24		1	5	15	40	65	1		5	15	36.5	60	60	1		5	20	45	70	70
GPT-5.5 · 2026-04-23		0.1	2.5	12	24.5	45	0.1		2.5	11.5	23	42	42	0.2		6	21	37	58	58
Claude Opus 5 · 2026-07-24		0.1	1	5	10	20	0.1		1	3.5	7	14	14	0.2		2.8	9.5	17.5	30	30
Qwen3.8 27B · 2026-08-14		0.1	1	5	13.5	30	0.1		1	5	15	30	30	0.4		3	9	23.5	45	45

P versus NP		2026	2030	2035	2040	2050	Riemann hypothesis		2026	2030	2035	2040	2050	Yang-Mills existence and mass gap		2026	2030	2035	2040	2050
GPT-4 · 2023-06-13		—	—	—	—	—	—		—	—	—	—	—	—		—	—	—	—	—
Qwen 72B · 2023-11-30		—	—	—	—	—	—		—	—	—	—	—	—		—	—	—	—	—
GPT-4o · 2024-08-06		1	5	10	20	35	1		5	15	27.5	50	50	1		5	15	30	45	50
Qwen2.5 72B · 2024-09-19		10	20	35	50	70	15		25	45	60	80	80	7.5		17.5	32.5	50	70	70
o1 · 2024-12-17		1	3	8	15	30	1		3	8	15	32.5	32.5	1		4	12	26.5	40	50
R1 Distill 32B · 2025-01-20		—	—	—	—	—	—		—	—	—	—	—	—		—	—	—	—	—
o3 · 2025-04-16		0.45	2.5	6.5	12.5	25	0.5		2	6	12	25	25	1		4	12	24	37.5	40
Qwen3 32B · 2025-04-29		2	10	15	25	40	1		5	10	20	32.5	32.5	1		5	10	22.5	37.5	37.5
GPT-5 · 2025-08-07		0.2	1.5	6	12	25	0.2		1.2	6	15	30	30	0.3		2	9	20	41	41
Claude Opus 4.5 · 2025-11-24		0.1	0.75	2.5	5.8	12	0.1		1.2	4	8.8	16	16	0.2		1.5	5	11	22	22
Claude Opus 4.6 · 2026-02-05		0.05	0.5	2	4.5	8	0.1		2	6	12	25	25	0.1		1	4	10	18	18
Qwen3.5 27B · 2026-02-24		0.35	3	12	30	47.5	1		5	15	35	60	60	1		5	15	40	65	65
GPT-5.5 · 2026-04-23		0.1	2.5	11.5	22.5	42	0.2		3.8	14	28	50	50	0.1		3	13	26.5	46.5	46.5
Claude Opus 5 · 2026-07-24		0.07	0.7	2.8	5.5	12	0.1		2	7	13	24	24	0.1		1.5	6	12.5	24.5	24.5
Qwen3.8 27B · 2026-08-14		0.2	1.8	6	15	32.5	0.2		2	8	15	35	35	0.2		1.8	8	19	35	35

Median across valid prompt/sample responses; n = 25–30 per observed model–problem pair (2,132 valid responses total).

— No valid recorded forecast (18 of 90 model–problem pairs). Unavailable: GPT-4. Pending: Qwen 72B, R1 Distill 32B.